

Weld Chemistry Prediction Performance Report (4.8 mm Electrode)

1. Model Validation Summary

Element	Best Model	R ²	RMSE	MAE	Validation Rating
Al	Linear Regression	0.149	0.0988	0.0834	Poor
C	SVR (RBF)	-0.208	0.0283	0.0188	Poor
Cr	SVR (RBF)	-0.370	1.9154	1.0839	Poor
Mn	ElasticNet	0.406	0.1658	0.0956	Fair
Mo	KNN	0.765	0.0803	0.0672	Good
Nb	Ridge	0.478	0.5158	0.4327	Moderate
Ni	Gaussian Process Regression	0.923	0.5030	0.4573	Excellent
P	Decision Tree	0.019	0.0069	0.0048	Poor
S	Linear Regression	-0.529	0.0094	0.0060	Poor
Si	Linear Regression	0.302	1.0542	0.5980	Fair
Ti	Linear Regression	0.703	0.2566	0.1942	Good

2. Test Sample Prediction Analysis

Element	Best Model	Actual (%)	Predicted (%)	Absolute Error	Relative Error (%)	Residual (Actual – Predicted)	Validation R ²	Prediction Quality
C	SVR (RBF)	0.037	0.049000	0.0120	32.43	-0.0120	-0.208	Poor
Si	Linear Regression	3.380	1.282429	2.0976	62.06	2.0976	0.302	Poor
Mn	ElasticNet	0.560	0.230630	0.3294	58.82	0.3294	0.406	Poor
Cr	SVR (RBF)	14.020	17.838754	3.8188	27.24	-3.8188	-0.370	Poor
Ni	Gaussian Process Regression	51.100	50.524673	0.5753	1.13	0.5753	0.923	Excellent
Mo	KNN	2.870	2.890698	0.0207	0.72	-0.0207	0.765	Excellent
Ti	Linear Regression	0.150	0.138396	0.0116	7.74	0.0116	0.703	Good
Al	Linear Regression	0.030	0.077142	0.0471	157.14	-0.0471	0.149	Very Poor

Nb	Ridge	3.350	2.817877	0.5321	15.88	0.5321	0.478	Fair
S	Linear Regression	0.021	0.003126	0.0179	85.11	0.0179	-0.529	Very Poor
P	Decision Tree	0.020	0.007000	0.0130	65.00	0.0130	0.019	Very Poor

3. Prediction Quality Summary

Quality	Criterion	Elements
Excellent	Error < 2%	Ni, Mo
Good	Error 2–10%	Ti
Fair	Error 10–20%	Nb
Poor	Error >20%	C, Si, Mn, Cr, Al, S, P

4. Element-wise Interpretation

Element	Remarks
C	Carbon is overpredicted by ~32%. The SVR model does not generalize well for this element, consistent with its negative validation R ² .
Si	Silicon is significantly underpredicted (62% error), indicating that the current formulation features do not adequately explain silicon recovery.
Mn	ElasticNet captures the overall trend but substantially underestimates manganese for this sample.
Cr	Chromium is overpredicted by 3.82 wt.%, reflecting poor model generalization and suggesting chromium recovery depends on additional process variables not present in the dataset.
Ni	Excellent agreement between prediction and experiment (1.13% error). Gaussian Process Regression successfully captures the relationship between formulation and nickel content.
Mo	Most accurate prediction (0.72% error). KNN performs very well for molybdenum in this dataset.
Ti	Good prediction with only 7.74% relative error, consistent with the model's good validation performance.
Al	Aluminum is considerably overpredicted. Its low concentration and limited training data likely reduce model accuracy.
Nb	Ridge Regression provides a reasonable estimate with a moderate error of 15.88%.
S	Sulfur is strongly underpredicted. Trace impurity elements are difficult to model accurately from formulation alone.
P	Phosphorus prediction deviates substantially, indicating insufficient information in the current dataset to model this impurity reliably.

5. Overall Performance Summary

Metric	Result
Total Elements Predicted	11
Excellent Predictions	2 (Ni, Mo)
Good Predictions	1 (Ti)
Fair Predictions	1 (Nb)
Poor Predictions	7
Best Validation Model	Gaussian Process Regression (Ni, $R^2 = 0.923$)
Most Accurate Test Prediction	Mo (0.72% error)
Second Most Accurate Prediction	Ni (1.13% error)
Third Most Accurate Prediction	Ti (7.74% error)

Conclusion

The machine learning workflow successfully identified different regression models for individual weld chemistry elements. **Gaussian Process Regression** achieved the strongest overall performance for **Nickel**, while **KNN** produced the most accurate prediction for **Molybdenum**. **Titanium** was also predicted with good accuracy using Linear Regression.

However, the models exhibited limited predictive capability for **Carbon, Silicon, Manganese, Chromium, Aluminum, Sulfur, and Phosphorus**, as indicated by large test errors and low or negative validation R^2 values. This suggests that the current dataset is sufficient to model elements that are directly influenced by alloy additions (e.g., Ni and Mo), but additional experimental samples and richer metallurgical descriptors (such as flux basicity, alloy recovery efficiency, or welding process parameters) would be required to improve prediction accuracy for impurity elements and more complex alloying behaviors.